

What is Machine Learning

In the real world, we are surrounded by humans who can learn everything from their experiences with their learning capability, and we have computers or machines which work on our instructions. But can a machine also learn from experiences or past data like a human does? So here comes the role of Machine Learning.

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In [ ]: # Making Machine Intelligent
```

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In [ ]: Machine Learning----

Machine---- Computer / System
Learning---- Training

Provide Data----Trained Model-----Predict Value--Target

Machine Learning--10-12

data----- prediction of some amount ,prediction of chances
```

Introduction to Machine Learning

A subset of artificial intelligence known as machine learning focuses primarily on the creation of algorithms that enable a computer to independently learn from data and previous experiences. Arthur Samuel first used the term "machine learning" in 1959. It could be summarized as follows:

Without being explicitly programmed, machine learning enables a machine to automatically learn from data, improve performance from experiences, and predict things.

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In [ ]: # There are Several Algos-----

Regression---prediction of price-----
Classification--- chances
```

```
In [ ]: Algo--- Stepwise Solution Of the Problem

sandeep write a code for swapping the values
```

```
In [ ]: kelvin---startup---4years---data---comes to archna for on the behalf of data my fut
startup

Studied the data---trained algo-----testing it on some likelihood---predict values
```

Classification of Machine Learning

At a broad level, machine learning can be classified into three types:

Supervised learning Unsupervised learning Reinforcement learning

Supervised Learning

In supervised learning, sample labeled data are provided to the machine learning system for training, and the system then predicts the output based on the training data.

The system uses labeled data to build a model that understands the datasets and learns about each one. After the training and processing are done, we test the model with sample data to see if it can accurately predict the output.

The mapping of the input data to the output data is the objective of supervised learning.

Supervised learning can be grouped further in two categories of algorithms:

Classification----prediction of chances Regression-----prediction of amount or some certain

Unsupervised Learning

Unsupervised learning is a learning method in which a machine learns without any supervision.

The training is provided to the machine with the set of data that has not been labeled, classified, or categorized, and the algorithm needs to act on that data without any supervision. The goal of unsupervised learning is to restructure the input data into new features or a group of objects with similar patterns. Unsupervised learning is a learning method in which a machine learns without any supervision.

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Clustering Association

1. info()---- the logical structure of our dataset

Column_Name not Null values data_type

2. Treat Null Values

can we drop null value--yes/no

3. categorical ---text values
numerical----- numbers---

4. encoding

"male":0,"feamle":1---

finding outliers

one of the friends of abhilsh showing an abnormal behavior

input
output

scaling ---- standard Scaling
MinMax Scaling

ML-----3 min

In []:

In []: